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# Operational Improvements in Sour TEG Dehydration Units

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## Abstract

One of the largest sour oil and gas processing stations in the south of Oman, operated by Petroleum Development Oman (PDO), was commissioned in mid-2019. The facility employs two triethylene glycol (TEG) dehydration units designed to reduce the water content in natural gas to required levels before it is injected back into the field for enhanced oil recovery. However, shortly after the plant became operational, both TEG units began to experience major operational challenges. These included high total suspended solids (TSS), low glycol pH, and excessive TEG losses. These problems led to frequent process disruptions, increased downtime, and raised significant health, safety, and environmental (HSE) concerns.

This study aims to systematically identify and assess the recurring performance issues in the TEG systems. A comprehensive root-cause analysis was conducted, followed by evaluation of practical and technically sound remediation measures. The study follows good engineering practices, with specific alignment to Shell's Glycol Type Gas Dehydration Design Engineering Practices (DEP), in addition to special considerations for sour service applications.

Key findings highlight the corrosive nature of acid gases such as hydrogen sulfide ($H_2S$) and carbon dioxide (CO2) dissolving into the glycol, which accelerated internal corrosion of carbon steel components. Additionally, it was observed that the flash vessel design was inadequate, allowing hydrocarbon-rich glycol to reach the contactor and negatively impact its performance. Further root causes included poor filtration upstream of the contactor, which permitted solids to accumulate, and process instability triggered by suboptimal pigging procedures and control valve behavior.

Several base recommendations were developed, tested, and successfully implemented on the dehydration units. These included switching to sodium acetate buffering to maintain a more neutral and stable glycol pH, upgrading the flash vessel internals with adjustable weirs and hydrocarbon skimmers, relocating the glycol make-up point to the flash drum to reduce contamination, installing cartridge filters at the TEG inlet scrubber to improve filtration efficiency, and enhancing process control through better flow regulation and pigging velocity limits.

**Keywords:** Sour Gas, TEG Dehydration, Corrosion, Acid Gas, Flash Vessel Design, Sodium Acetate, Operational Improvement, Sour Gas Processing, Glycol Losses

## Introduction

Acid gas is a combination of carbon dioxide ($CO_2$) and hydrogen sulfide ($H_2S$), often saturated with water is commonly generated as a by-product during the sweetening of sour gas (Peric, 2019). A widely applied solution for handling this unwanted by-product is acid gas injection, a method recognized for being both economically favorable and environmentally responsible. In this process, the acid gas is compressed to a sufficiently high pressure and injected into a disposal well for permanent storage.

The presence of water in acid gas poses significant operational challenges. It contributes to internal corrosion, hydrate formation, and reduced equipment efficiency. One must remove this water prior to injection to ensure safe and effective disposal. Although dehydration via compression and cooling is essential for acid gas, water absorption using triethylene glycol (TEG) remains the most popular and effective method for achieving highly dry gas streams (Grynia, Carroll & Griffin, 2010).

TEG is favored due to its low vapor pressure, relatively low energy cost for pumping, and efficient dehydration performance. It also offers a high affinity for water and regenerates easily to a high purity typically at around 400 °F and can achieve strong dew point depression (often exceeding 90 °F). While other glycols such as diethylene glycol (DEG), monoethylene glycol (MEG) and triethylene glycol (TREG) exist, TEG provides a superior balance of dehydration efficiency, operational cost, and reliability.

The field of gas dehydration has seen ongoing innovation and optimization (Petropoulou, 2019) with research focusing on improving process quality and reducing costs. Despite various techniques emerging, absorption using TEG remains the industry standard for achieving low water content in gas while minimizing environmental impact and operational expenditure (Dakhil, 2012). However, TEG-based systems, particularly in sour gas service, introduce additional risks such as accelerated corrosion, foaming, increased glycol loss, solid fouling, and safety concerns. Traditional designs tailored for sweet gas often fail to fully address these challenges.

Glycols, chosen for their hygroscopic nature and favorable physical properties—such as low viscosity and non-corrosiveness—have made TEG the most widely used liquid desiccant in dehydration systems. Other glycols like DEG, MEG, and TREG serve specific niche roles, such as hydrate inhibition or low-temperature dehydration, but cannot match TEG in overall applicability and efficiency.

At our facility, two inline TEG dehydration trains were designed to meet a strict output specification of <40 mg/Sm$^3$ water in the dry gas. Though the units were successfully commissioned in mid-2019 and initially met this specification, operational issues soon emerged. Filters at TEG inlets clogged within 20 hours, the pH of lean glycol dropped to 6 or below, glycol makeup volumes surged to 120m3/day—well above the design target of ~75 m3/day—while flash vessel foaming and hydrocarbon carryover caused intermittent regeneration failures. These issues reduced plant uptime from over 98% to approximately 92%.

Existing practices—such as using Petromeen antifoam agents, standard high-efficiency mist mats, and conventional contactor packing—proved inadequate, and staff lacked access to sour-gas- specific TEG troubleshooting literature.

This study is designed to comprehensively assess the system by evaluating the extent and distribution of deposits forming throughout, identifying the underlying root causes of these deposits, and developing practical mitigation strategies. The assessment includes both qualitative and quantitative analyses of deposit characteristics, operational conditions contributing to their formation, and potential process improvements to minimize future deposition and maintain optimal system performance.

Consequently, a comprehensive internal review was launched with three key objectives:

1. **Identify root causes** of frequent TEG filter blockage, glycol acidity, elevated TEG losses, and control instability.
2. **Benchmark performance issues** against Shell's Design Engineering Practices (DEP) and established sour-service guidelines.

3. **Develop and pilot engineering solutions** to enhance reliability, efficiency, and safety of sour gas TEG dehydration units.

## System Overview and Sour-Service Challenges

Figure 1 illustrates a typical Triethylene Glycol (TEG) system process flow scheme, which is commonly used for natural gas dehydration. The diagram outlines the main components and flow sequence, including the gas inlet, contactor column, glycol circulation loop, regeneration unit, and associated heat exchangers. This schematic provides a simplified overview of how wet natural gas is treated with TEG to remove water vapor, ensuring pipeline specifications are met and preventing hydrate formation and corrosion.

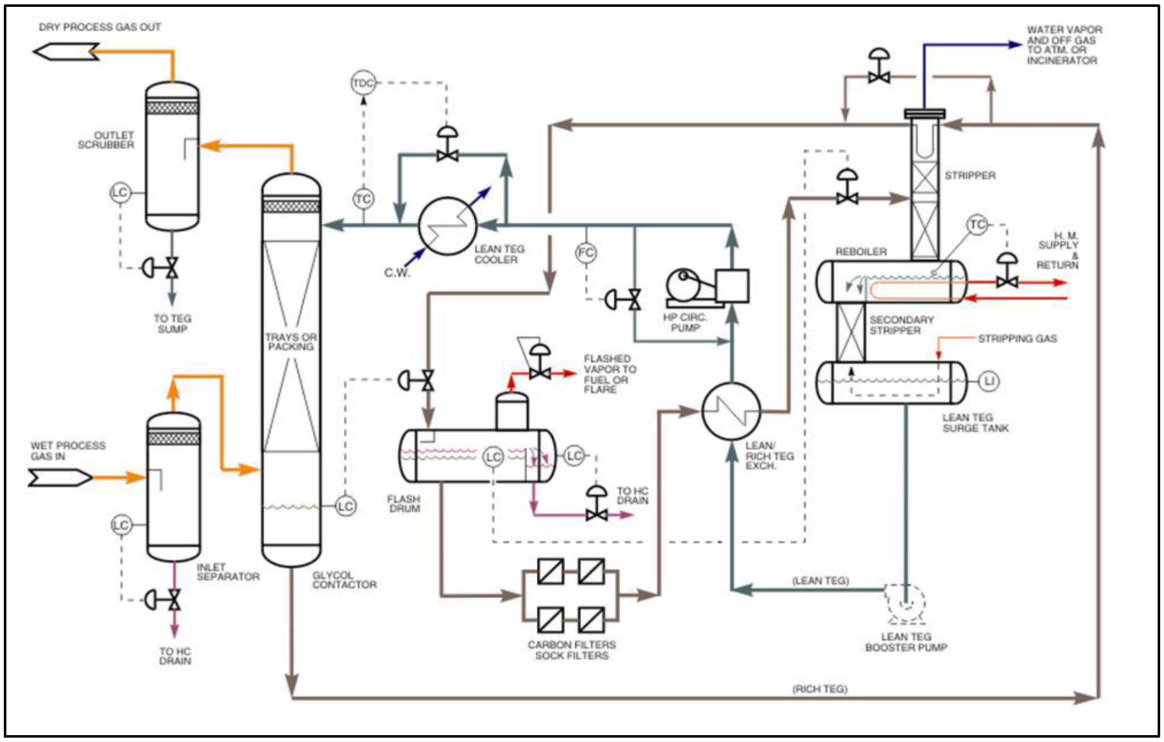


**Figure 1—Typical TEG System Process Flow Scheme (Dakhil, 2012)**

Table 1 provides a comprehensive summary of the main findings from the root cause analysis, highlighting the underlying causes of each issue and their respective operational or safety impacts. This overview enables a clearer understanding of the relationships between identified problems and their effects on system performance, guiding the development of targeted mitigation strategies.

**Table 1—Root cause analysis of the various identified issues**

| Issue | Cause | Impact |
|---|---|---|
| **Low TEG pH High Corrosion Rate** | Presence of sour hydrocarbons ($H_2S$, $CO_2$) in the lean TEG stream reacts with water to form acidic species, lowering the TEG pH and accelerating corrosion. | Increased corrosion in pipelines, heat exchangers, and the reboiler can lead to equipment damage, reduced lifespan, potential leaks, and safety hazards. |
| **High TSS Frequent Filter Replacement** | Accumulation of iron sulfide, elemental sulfur deposition, and carbon carryover from the gas stream causes elevated Total Suspended Solids (TSS) in the system. | Filters become clogged rapidly, leading to frequent replacements, increased maintenance downtime, higher operating costs, and potential process interruptions. |
| **High TEG Losses** | Sudden ramp-up or ramp-down of gas compressors creates unstable flow conditions that can carry glycol out of the system. | Increased TEG consumption, higher operational costs, potential environmental release, and reduced efficiency of the dehydration process. |

| Issue | Cause | Impact |
|---|---|---|
| **High HC Content in Lean TEG** | Inefficient hydrocarbon skimming in the flash gas drum allows hydrocarbons to remain in the lean TEG. | Contamination of the TEG system, reduced dehydration efficiency, potential foaming issues, and compromised downstream processing operations. |

## Methodology

To achieve the study's objectives, a systematic approach was employed, incorporating both qualitative and quantitative methods. The methodology consisted of the following key steps

- **Data Collection for Performance Evaluation:** Operational data, including TEG flow rates, pH levels, hydrocarbon content, and system losses, were collected and analyzed to evaluate system performance and identify deviations from design parameters.
- **Site Inspections and Observations for Corrosion and Material Analysis:** On-site inspections were conducted to assess the condition of pipelines, heat exchangers, pumps, and other critical equipment. Material samples were analyzed to detect corrosion, deposit formation, and other degradation mechanisms.
- **Root Cause Analysis (RCA) with Vendor Reviews:** A detailed RCA was performed to determine the underlying causes of observed issues, supported by vendor consultations and reviews of equipment design, operational practices, and maintenance records.
- **Lab Experiments to Maintain TEG Quality:** Laboratory tests were carried out to monitor and maintain TEG chemical composition, pH, and contaminant levels, ensuring optimal dehydration performance and chemical stability under operating conditions.

## Results and Discussion

Figure 2 presents a comparison between a fresh TEG sample and used TEG collected from the two dehydration units. The used TEG shows noticeable darkening and turbidity compared to the clear fresh sample, indicating the presence of accumulated solids, hydrocarbons, and other contaminants. These visual changes reflect chemical degradation, water saturation, and potential buildup of corrosion products, emphasizing the importance of regular monitoring, filtration, and regeneration to maintain dehydration efficiency and prevent operational issues.

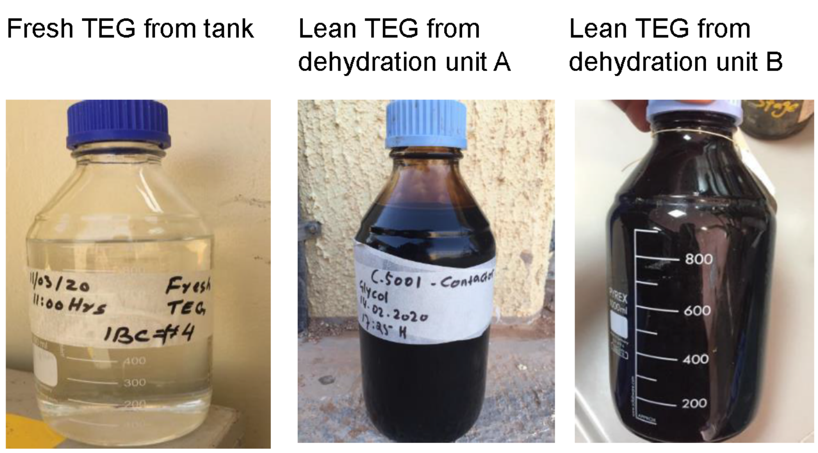


**Figure 2—TEG samples**

Fresh TEG from tank Lean TEG from Lean TEG from dehydration unit A dehydration unit B

It can be observed that the fresh TEG sample is clear and transparent while the samples collected from the two dehydration units are of black color indicating some form of contamination.

Figure 3 shows the solid sample collected from TEG dehydration knock out drum (KOD) inlet. The inlet KOD after inspection was found fully choked with solid particles and elemental sulfur. Around 3 to 5 tons of Sulfur was collected from one of the unit's inlet KOD.

**Figure 3—Solid sample collected from inlet KOD**

The Glycol Circulation Pump section, including the strainer and mechanical filters, was thoroughly inspected. The pump section strainer was found to be completely clogged with solids and oily materials, as illustrated in Figure 4. This blockage indicates significant accumulation of contaminants, which can impede pump performance, reduce system efficiency, and potentially cause operational issues if not addressed.

**Figure 4—Glycol pump suction strainer and mechanical filters**

The cut-outs of the filter medium were collected and sectioned into an upstream and downstream layer and all samples were dried. All samples were investigated by optical microscopy. Figure 5 shows the light optical microscopic image of a section of a filter element. The light microscopy confirmed the visual impression. Both the upstream- and downstream sides were observed to be contaminated by a black substance. Further the contamination on the filter medium was observed to be most intensively adhering to the filter medium on the upstream side and less intense on the downstream side. The upstream side of the filter medium presented a black layer covering the entire surface (Figure 5).

Figure 5—Overview of filter medium (left upstream side, right: downstream side)

The results for Scanning electron microscopy (SEM) and energy dispersive X-ray (SEM/EDX) investigation of all samples are given below in figure 6 and 7. Observations from SEM investigations support the findings from optical light microscopy. A distinct coating was visible overall on all upstream side samples (Figure 3). The downstream side samples showed the contamination load was significantly lower in comparison to the upstream side samples.

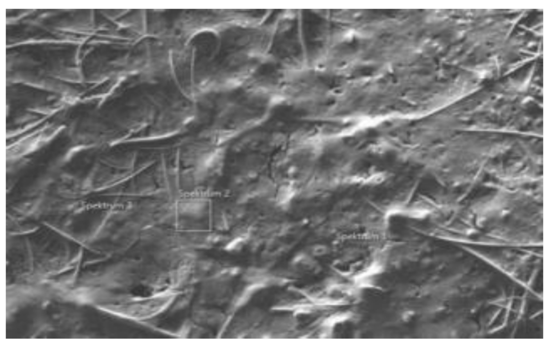

Figure 6—Scanning Electron Microscopy (SEM) Overview

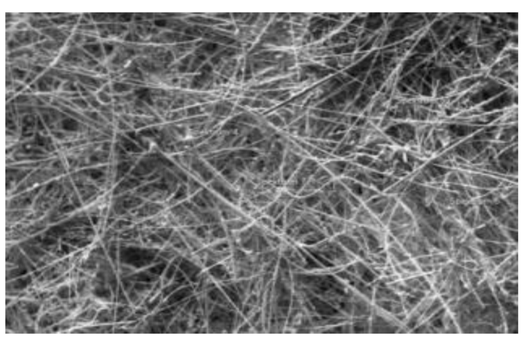

Figure 7—SEM overview of HFU downstream side

Figure 8 shows the EDX spectrum of contamination and all investigated particles in the EDX spectrum of the filter section. Sharp peaks can be seen for C, O and S differing in their intensities and composition. Presence of carbon indicates high liquids that are most probably "viscous" hydrocarbons that have also participated in fouling the filters. High content of Sulfur further indicates presence of Sulfur in the TEG.

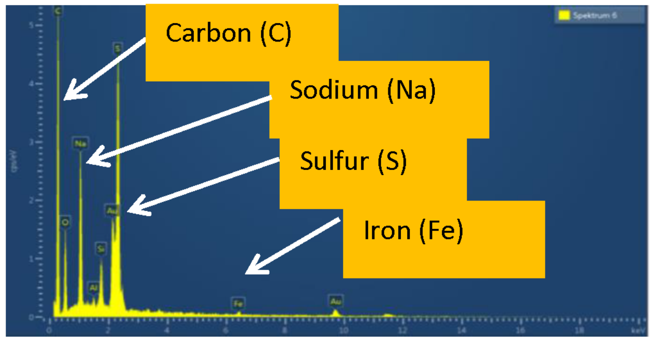


**Figure 8—Energy Dispersive X-ray (EDX) spectrums of Filter**

## Effect of sodium acetate on the pH of lean Glycol

The effect of sodium acetate on the pH value of lean glycol was investigated. For 100 ml of glycol used in this experiment collected from TEG unit the sample pH was 6.81 ppm. Sodium acetate was injected gradually (0.5 ml each time) and the change in pH was noted. The experimental results are shown in figure 9. Sodium acetate proved to be very effective in controlling the pH of glycol in the range of 6.5 to 8. Therefore, no pH injection is required to correct the current pH.

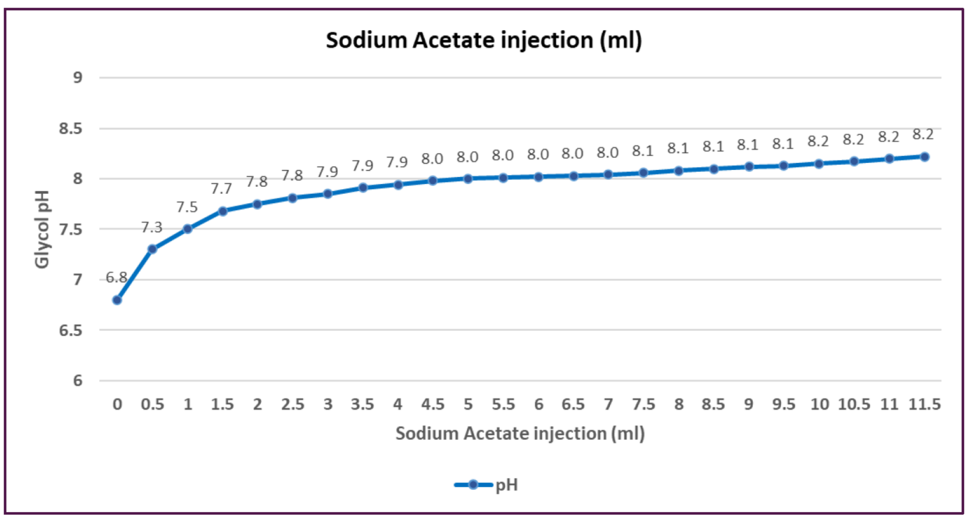


**Figure 9—Effect of sodium acetate on lean Glycol pH**

## TEG Losses

Every dehydration unit is prone to glycol loss; it is part of everyday operation. Loss in glycol occurs due to entrainment or vaporization. The physical loss of glycol is probably the most important operating problem in the dehydration system. An acceptable glycol loss for a properly designed and maintained dehydration system is 20-60 L/MMSCM of gas in accordance with shell standards. The glycol loss as per design for the two dehydration units under study is 30 L/MMSCM. Figures 10 and 11 clearly indicate the real TEG losses from both dehydration units against the limit losses on both the TEG units respectively.

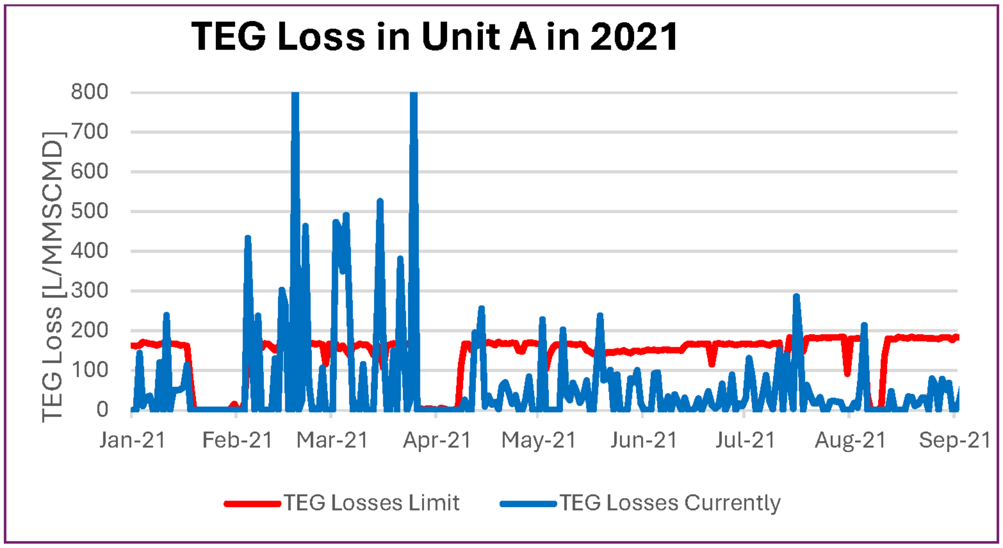


**Figure 10—TEG Loss in Unit A in 2021**

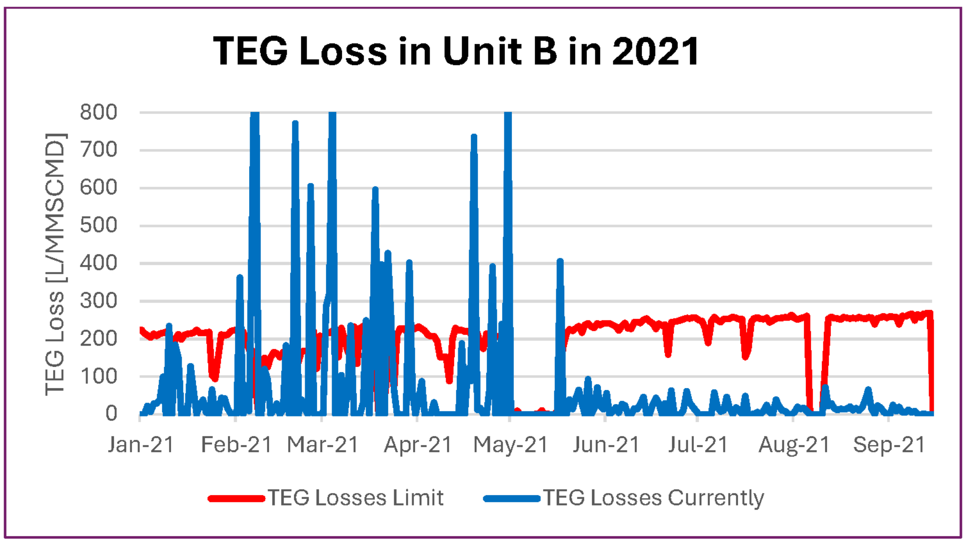


**Figure 11—TEG Loss in Unit B in 2021**

## High Moisture Content Management in TEG Dehydration Unit

During routine monitoring of one of the dehydration units, an increase in the moisture content of the gas outlet was observed. This rise was primarily attributed to a reduction in the Triethylene Glycol (TEG) circulation rate, a measure implemented to prevent the surge drum from tripping due to low liquid level (LL).

The low liquid level in the surge drum was suspected to be caused by a choke in the stripping column and/or the lean side of the lean-rich heat exchanger. Additionally, higher contactor inlet gas temperatures during daytime operations contributed to the increased moisture content in the dry gas.

## Proactive steps adopted to overcome the challenges

To address the challenges of high moisture in TEG and high TEG loss, the following proactive steps were taken until upcoming turnaround:

1. **Temperature Control Post-Cooling:**
    - The first-stage aftercooler for the high-pressure acid gas (HP AG) is maintained at 60°C.

- The second-stage aftercooler is maintained at 57°C.

2. **Adjustment of Acid Gas Outlet to Dehydration Unit Temperature:**
   - The gas outlet temperature from the Acid Gas Compressors (AGC) was proactively adjusted to maintain moisture content below 30 ppm, even with the reduced TEG circulation rate of 7,000 kg/hr (with a trip setpoint at 6,200 kg/hr).
3. **Operational Adjustments in Case of High Moisture Content:**
   - If moisture levels continue to rise, gas flow from the AGC to Unit A TEG dehydration inlet FCV will be reduced from 90% to 50% valve opening.
   - Simultaneously, flow to the B Dehydration Unit will be increased by adjusting inlet FCV accordingly.
4. **Regular Maintenance:**
   - Filter elements are proactively replaced biweekly. To prevent bypassing, the 2″ drain valves are ensured to be closed after replacing filter elements.

These measures are designed to ensure that the dry sour gas sent to the MGI Compression remains within specification (below 40 mg/Sm³) until the upcoming turnaround.

## Operational Improvement in Sour TEG System

It is also suggested to consider the following points (shown in figure 12) during the design of sour service TEG dehydration unit:

1. Install Upstream Cartridge Filter.
2. Modify the flash vessel to have skimming bucket for HC or extend the height of weir.
3. Provide Filter after Activated Carbon Filter.
4. TEG Makeup to be routed to Flash vessel.
5. Provide Standby L/R Exchanger with solation Valve around L/R Exchanger.
6. Provide Standby Reboiler Bundle.
7. Provide Online pH Analyzer.

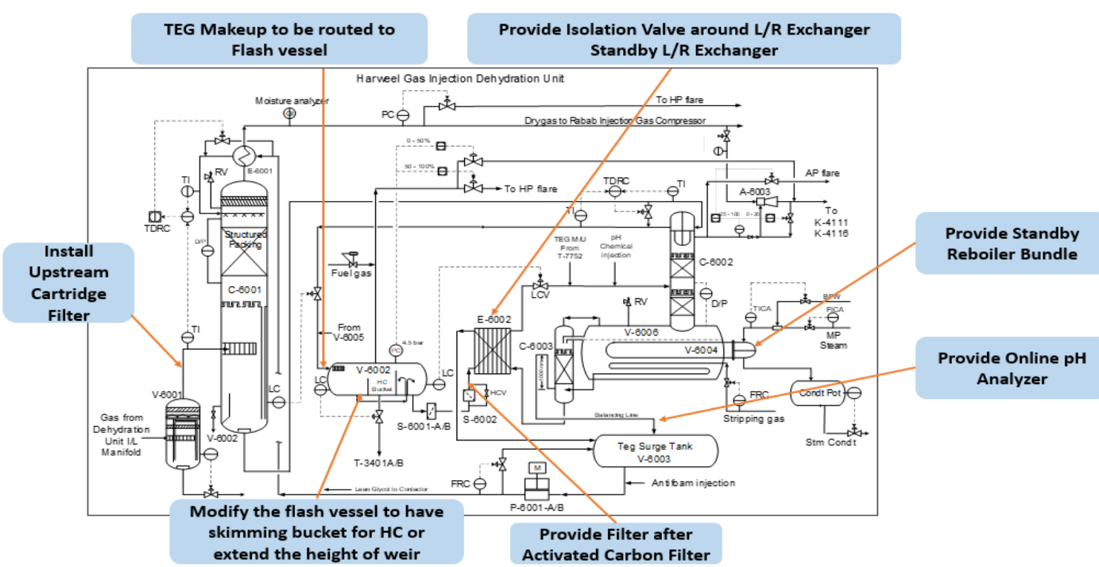


**Figure 12—Sour TEG Unit Further Design Improvements Proposal**

## Conclusions

The study demonstrates several key improvements and operational insights for the TEG system:

- Sodium acetate is very effective chemical for sour TEG in controlling pH.
- After sorting out the elemental Sulfur deposition, mechanical filter replacement frequency reduced drastically.
- TEG losses were reduced within the design limit.
- The changes to the pH solution and dosing with sodium acetate have improved the stability and performance of the sour services.
- Addressing elemental sulfur deposition has effectively reduced operational issues with solids buildup.
- The adjustments to the pigging velocity and valve control have contributed to more stable operations by minimizing gas flow fluctuations.
- Additionally, the proposed flash vessel design modification and HC skimmer installation aim to further enhance system efficiency and reliability.

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